

John Ma

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Portfolio: jipperspippers.github.io/

Education

Honours BSc in Computer Science and Statistics

Specialist Program in Machine Learning and Data Science (Co-Op)

University of Toronto

3.5/4.0 GPA, Sep 2020 - Apr 2025

Experience

Software Engineer

Sep 2025 - Jun 2026

Uma Ltd.

- Architected an edge computer vision pipeline in C++ to optimize NPU inference, achieving ~ 28 ms end-to-end latency and outperforming standard benchmarks by 132%.
- Deployed 6 microservices designed for large-scale systems telemetry, integrating security software development practices with $\sim 99.9\%$ uptime across 4 months.
- Engineered custom low-level memory management and device interfaces in C++, eliminating serialization overhead and reducing end-to-end system latency by $\sim 30\%$ in resource-constrained environments.
- Accelerated test suite implementation from zero to $\sim 80\%$ test coverage, utilizing AI productivity tools (e.g., Antigravity, GitHub Copilot) to generate comprehensive functional, boundary, and regression tests.

Machine Learning Engineer

Apr 2025 - Aug 2025

Themis AI

- Implemented active learning and information retrieval algorithms across scalable ML pipelines, reducing scoring latency by 70% across 100k+ data samples.
- Developed scalable Python CLI tools accelerating configurable multi-annotator labeling workflows with a $\sim 30\%$ reduction in label noise.
- Reduced GPU costs by migrating from persistent instances to RunPod serverless instances, reducing monthly costs by 50%.

Machine Learning Engineer

Sep 2023 - Sep 2024

UTAT Space Systems

- Designed scalable Python data infrastructure and synthesis pipelines, generating 5,000+ hyperspectral images to support robust Computer Vision and Machine Learning model training.
- Diagnosed and resolved complex CUDA, GPU memory, and Python dependency conflicts across distributed systems, reducing data post-processing times by $\sim 50\%$.
- Co-authored peer-reviewed research presented at the SmallSat Conference 2024, demonstrating spatial-spectral attention models that yielded a 15% improvement in PSNR and 5% increase in SSIM.

Build Infrastructure Developer

Sep 2021 - Dec 2021

Blackberry QNX

- Migrated CI/CD infrastructure from static VMs to lightweight Docker containers, reducing host resource overhead by 95% and eliminating build queue wait times.
- Automated container documentation and workflow monitoring via Python and Bash scripts, eliminating 20 manual engineering hours per week while ensuring operational tracking.
- Engineered a custom Jenkins infrastructure plugin to enhance build reliability, achieving 99% uptime and cutting continuous integration failure rates by $\sim 40\%$.

Relevant Projects

jorblin | AI Resume Tailoring Platform (JavaScript, React, Node.js, Docker/Linux)

(Jun 2026)

- Built a full-stack web application end-to-end using a React/Vite client, Express REST API, and containerized Docker/nginx deployment on Linux, owning priorities as sole developer.
- Engineered a natural language processing (NLP) pipeline over multi-provider LLM APIs (Gemini, Claude, OpenAI) for keyword extraction and information retrieval scoring.

Beyond the Visible: Hyperspectral Image Denoising (Python, PyTorch, CUDA)

(Jun 2024)

- Developed a 3D diffusion architecture for joint spectral-spatial denoising with distributed GPU acceleration, published and presented at the SmallSat Conference 2024 (arXiv: 2406.10724).

Technical Skills

Languages: Python, C++, Java, JavaScript, C, TypeScript, Bash, SQL

Core Competencies: Data Structures, Algorithms, Distributed Systems, Machine Learning, Web Application Development, Mobile Application Development, Unix/Linux, Information Retrieval, Natural Language Processing, Parallel Systems, Networking, Accessible Technologies

Developer Tools & Platforms: Docker, Git, CI/CD (Jenkins, GitHub Actions), AWS, Google Cloud, Node.js, React, ROS, AI productivity tools (LLM tooling), PyTorch, scikit-learn, pandas